Think Cloud Think Azure!

Azure Core and Governance

Build • Secure • Govern • Scale

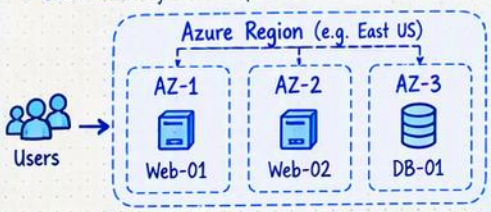
"Good Governance today = Fewer Incidents tomorrow" ☺

Remember:
- ☑ Understand
- ☑ Visualize
- ☑ Practice
- ☑ Explain
- ☑ You can do it!

1 Regions and Availability Zones

What it is: A Region is a geographic area with one or more datacenters. Availability Zones (AZs) are physically separate datacenters within a region, with independent power, cooling and networking.

Why it matters:
- Low latency for end users
- High availability and fault tolerance
- Data residency and compliance



Example: Deploy a web app across 3 AZs to handle datacenter failure.

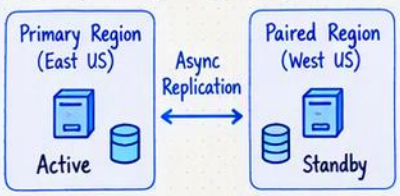
☆ **Remember:** Zones protect against datacenter failures.

⚠ **Interview trap:** An Availability Zone is **NOT** a separate region.

2 Region Pairs (Disaster Recovery)

What it is: Two regions within the same geography (Typically 300+ miles apart) paired for disaster recovery and business continuity.

Benefits:
- Protection against regional disasters
- Platform updates happen sequentially
- Used for geo-replication (e.g. GRS, ASR)
- Priority recovery in major outages



Example: Use Geo-Redundant Storage (GRS) to replicate data to the paired region.

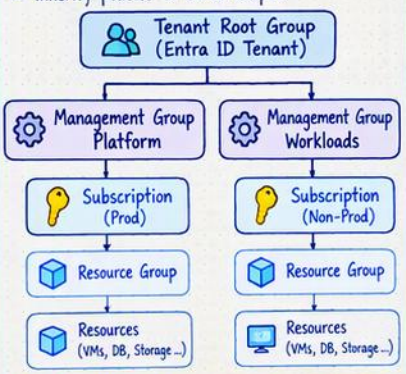
☆ **Remember:** Region pairs protect against regional (not datacenter) level disasters.

⚠ **Interview trap:** Not all regions are paired. Always confirm the exact pairing for the service.

3 Management Groups and Subscriptions

What it is: Management groups organize subscriptions. A subscription is a billing, quota and access boundary.

Why use it:
- Apply policy and RBAC at scale
- Separate environments (Prod / Non-Prod)
- Cost management and governance
- Inherity policies to child scopes



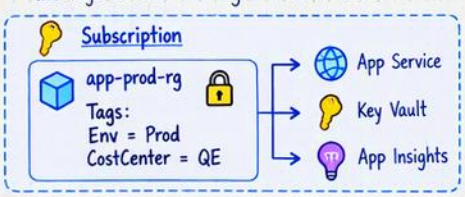
☆ **Remember:** Governance flows top-down.

⚠ **Interview trap:** A subscription is not the same as an Entra tenant. One tenant can have many subscriptions.

4 Resource Groups, Locks and Tags

What it is: A resource group is a container for Azure resources. Locks prevent accidental deletion or modification. Tags are key-value pairs for metadata (cost, ownership, etc.).

Why use it:
- Manage resources as a logical group
- Apply locks for protection
- Use tags for cost tracking and automation



Example: Create a resource group for your web app and add a CanNotDelete lock and tags.

☆ **Remember:** Tags do not automatically inherit.

⚠ **Interview trap:** Deleting a resource group can delete all contained resources. Recovery depends on backups/ soft delete and is not automatic.

5 Region vs Availability Zone

What it is: Region = geographic boundary. Zone = isolated datacenter within a region.

Feature	Region	Availability Zone
Scope	Geographical area (e.g. East US)	Datacenter within a region
Failure domain	Region-wide	Datacenter-level
Distance	Hundreds of miles (from pair)	< 2 ms latency (within region)
Purpose	DR, compliance, data residency	High availability
Example	East US, West US	AZ-1, AZ-2, AZ-3 (in East US)

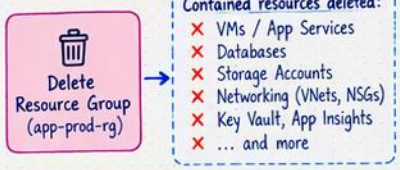
Use when: Choose region for proximity/compliance, and zones for resilience.

☆ **Remember:** HA inside a region: zones. DR across geography: multiple regions.

⚠ **Interview trap:** Zones are not a replacement for multi-region disaster recovery.

6 Deletion Impact (Resource Group)

What happens: Deleting a resource group can delete all contained resources.



Recovery: Use backups, soft delete (where available) or Infrastructure as Code (IaC) to recreate.

☆ **Remember:** Locks + least privilege + approval process + backups + IaC.

⚠ **Interview trap:** A lock does not replace backup. IaC recreates infrastructure but not necessarily data.

7 Locks vs RBAC

What it is: Locks prevent actions on resources. Consis who can perform actions.

	Locks	RBAC
Purpose	Block accidental deletion or modification	Manage access permissions
Scope	Resource, Resource Group, Subscription	Management Group, Subscription, Resource Group, Resource
Types	CanNotDelete (Delete)	Reader, Contributor, Owner, custom roles
Effect	Applies to all users (including Owner)	Allows or denies based on assigned role

Example: Use a CanNotDelete lock on production resource groups and grant only Contributor role to the deployment team.

8 Key Takeaways

- ☑ Choose the right region for user proximity and compliance.
- ☑ Use Availability Zones for high availability.
- ☑ Use region pairs for disaster recovery.
- ☑ Organize subscriptions with management groups.
- ☑ Use resource groups, locks and tags for better governance.
- ☑ Understand the difference between locks and RBAC.
- ☑ Always verify service availability and region pairing.
- ☑ Plan for deletion scenarios and have a recovery strategy.

Small Steps Big Learning ...

Design Secure Govern Repeat

"Well governed resources build resilient solutions"

Azure Networking – Core

Connect Securely • Design Clearly • Troubleshoot Confidently

Networking is the foundation that connects everything in Azure !

Same Network More Opportunities ☺

1 Virtual Network (VNet)

What it is: Logically isolated network in Azure. It's your private network in the cloud.

Why use it?

- Isolate resources
- Control traffic
- Connect to on-prem (VPN/ExpressRoute)
- Build multi-tier architectures

VNet 10.0.0.0/16

Example:

10.0.0.0/16 for Contoso app with web, app and db subnets.

Interview trap:

VNet is not the same as a subnet. It's a larger address space.

2 Subnet

What it is: A range of IP addresses within a VNet. Used to group resources and apply NSGs/UDRs.

Example:

- VNet: 10.0.0.0/16

Web subnet	10.0.1.0/24
App subnet	10.0.2.0/24
DB subnet	10.0.3.0/24

Notes:

- Each resource must be in a subnet.
- Subnets can't overlap.
- Some Azure services require dedicated subnets (e.g. AzureBastion, App Gateway, AKS).

Interview trap:

- You can't move a resource to a different subnet without redeploying (or supported operation).

3 Network Security Group (NSG)

What it is: Acts as a firewall for your subnet or NIC. Contains inbound and outbound security rules.

Example:

Allow HTTP (80) from Internet to Web subnet, deny all other inbound traffic.

NSG Rules (example)			
Priority	Source	Port	Action
100	Any	80	Allow
200	Any	443	Allow
300	Any	Any	Deny

Notes:

- Can be associated at subnet or NIC level.
- Rules are stateful (return traffic allowed).
- Use service tags (e.g. AzureCloud).

Interview trap:

- NSG is not an application firewall (it works at L3/L4, not HTTP/S).

4 Application Security Group (ASG)

What it is: Logical grouping of VMs/NICs. Use ASGs in NSG rules instead of individual IPs.

Why use it?

- Simplifies rule management
- Dynamic membership
- Useful in scalable environments (e.g. VMSS, AKS)

Example:

Create ASG "web-servers" and allow traffic from "app-servers" on port 8080.



Interview trap:

- ASG is not a load balancer and does not provide traffic distribution.

5 UDR (User Defined Route) & Azure Firewall

What it is: UDR lets you control routing of traffic from a subnet. Azure Firewall is a managed, stateful firewall used for central inspection and filtering.

Route Table (UDR)		
Address Prefix	Next Hop Type	Next Hop
0.0.0.0/0	Virtual Appliance	10.0.100.4
10.1.0.0/16	VNet	Local

Example:

Send internet traffic from spoke subnets to Azure Firewall.

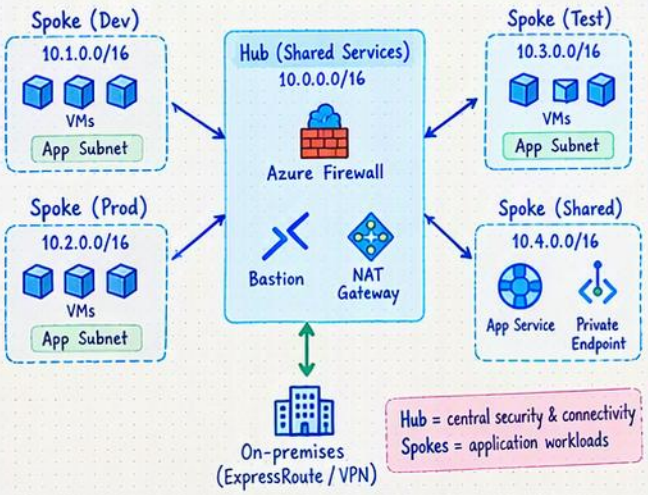
Notes:

- Associate route table with subnet.
- Azure Firewall provides application and network level filtering (FQDN, TLS).

Interview trap:

- UDR alone does not filter traffic. It only controls the route (next hop).

Hub-and-Spoke Architecture (Example)



Hub = central security & connectivity
Spokes = application workloads

6 Public IP & NAT Gateway

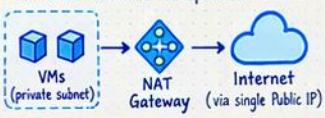
What it is: Public IP gives a resource a public address. NAT Gateway provides outbound internet access for multiple resources using a single public IP.

Use cases:

- Public IP – for internet-facing services (e.g. Load Balancer, App Gateway)
- NAT Gateway – outbound internet for private resources (e.g. VMs in subnet)

Example:

Use NAT Gateway for VMSS in private subnet to download OS updates.



Interview trap:

- NAT Gateway is for outbound only. It cannot be used for inbound connections.

7 Azure Bastion

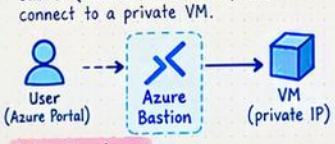
What it is: A secure, managed service to connect to VMs over RDP/SSH without public IP addresses.

Benefits:

- No public IP on VMs
- Access via Azure portal (or native client)
- Integrated with Entra ID and RBAC

Example:

Deploy Bastion in a dedicated subnet (AzureBastionSubnet) and connect to a private VM.



Interview trap:

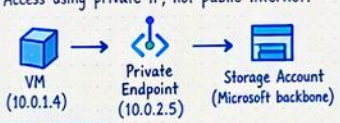
- Bastion requires its own dedicated subnet (AzureBastionSubnet) with minimum /26.

8 Private Link & Private Endpoint

What it is: Private Link provides private connectivity to Azure PaaS services over the Microsoft backbone. A Private Endpoint is a private IP in your VNet for that service.

Example (Storage):

Create a private endpoint for a storage account. Access using private IP, not public Internet.



DNS Example:

Private DNS zone: privatelink.blob.core.windows.net
Record: mystorage.privatelink.blob.core.windows.net → 10.0.2.5

Interview trap:

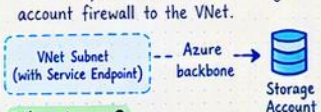
- You need private DNS for name resolution. Otherwise traffic may go to public endpoint.

9 Service Endpoint

What it is: Extends your VNet identity to Azure services (e.g. Storage, SQL) over the Azure backbone. No private IP is created (unlike Private Endpoint).

Example:

Enable service endpoint for Storage on subnet, then restrict the storage account firewall to the VNet.



When to use?

- Simple and less complex than Private Link
- Good for data exfiltration prevention
- No private IP, uses service endpoint

Interview trap:

- Service Endpoint does not provide a private IP. For full private connectivity, use Private Endpoint.

10 Key Takeaways

- ☑ Plan your VNet and subnet design early.
- ☑ Use NSGs and ASGs for granular security.
- ☑ Control routing with UDRs and centralize inspection with Azure Firewall.
- ☑ Use NAT Gateway for outbound internet access.
- ☑ Use Azure Bastion for secure VM access (no public IP).
- ☑ Use Private Link for true private connectivity to PaaS.
- ☑ Use Service Endpoints for simpler VNet to service connectivity.
- ☑ Design hub-and-spoke for scale and governance.

"A well-designed network makes your Azure environment secure, scalable and future-ready!" ☺

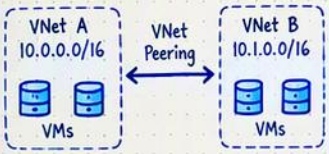
Azure Networking – Hybrid and Troubleshooting

Connect On-Prem, Cloud and Solve Network Issues with Confidence!

Network Well
Troubleshoot Smart
Build Resilient Solutions !

1 VNet Peering

What it is: Connects 2 VNets privately over Microsoft backbone (low latency, high bandwidth).



Use when:
- Connect VNets within same/diff region.
- App tiers across VNets.
- Hub-spoke architecture.

Example:
Peer spoke VNet (app) with hub VNet (shared services).

Interview trap:
Peering is not transitive.
If A↔B and B↔C, A cannot reach C without direct peering (or use a hub).

2 VPN Gateway

What it is: Secure, encrypted connection between on-prem network and Azure over the internet (IPsec/IKE).



Use when:
- Site-to-site (branch/office to Azure)
- Point-to-site (individual users)
- Cost-effective option

Example:
Connect office network (192.168.1.0/24) to Azure VNet (10.0.0.0/16) using VPN Gateway.

Interview trap:
Not suitable for very high bandwidth or ultra-low latency, Consider ExpressRoute for that.

3 ExpressRoute

What it is: Private, dedicated connection from on-premises to Azure via a service provider (not over internet).



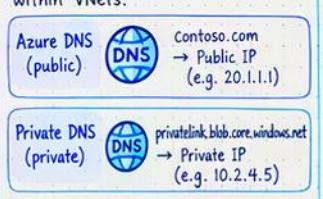
Benefits:
- More reliable, consistent performance
- Higher bandwidth (up to 100 Gbps+)
- Avoids internet, lower latency
- Supports private peering (VNet), Microsoft peering (O365), public peering

Example:
Use ExpressRoute to connect datacenter to Azure for SAP workloads.

Interview trap:
ExpressRoute does not encrypt traffic (by default). Use IPsec over ExpressRoute if encryption is required.

4 Azure DNS & Private DNS

What it is: Azure DNS hosts public DNS zones. Private DNS provides name resolution for private IPs within VNets.



Use when:
- Host custom domain in Azure
- Enable private name resolution for private endpoints
- Hybrid DNS with on-prem (conditional forwarders)

Example:
Create private DNS zone privatelink.sql.azure.com and link it to your VNet.

Interview trap:
Private DNS zone must be linked to the VNet. Without it, name resolution fails.

Troubleshooting Network Issues – General Flow

Follow this order! It saves time



Useful Commands:
- nslookup <name>
- Test-NetConnection <ip> -Port 443
- az network watcher test-connectivity ...
- az network nic show-effective-nsg ...
- az network route-table show ...
- az vpn-gateway show ...
- az network private-endpoint show ...
- az network watcher troubleshooting start ...

Check one layer at a time

Common Tools & Services

- Azure Network Watcher (Connection troubleshoot, IP flow verify)
- NSG Flow Logs (check allowed/denied traffic)
- Effective Routes (see actual routing)
- Packet Capture (VM/NVA)
- Azure Monitor & Log Analytics (diagnose)
- Azure Firewall logs
- DNS query tools (nslookup, dig)
- Service health (Azure status)
- Use Azure CLI / Portal / PowerShell

→ Observe
→ Identify
→ Fix
→ Verify
→ Document

"Most network issues are configuration issues → follow the flow, don't guess!"

5 VM has no Internet

Symptoms:
- Cannot browse internet
- apt/yum update fails
- Can reach other VMs in VNet

Common causes:
- No public IP
- No default route (0.0.0.0/0)
- UDR sending traffic to NVA
- NSG blocking outbound
- Firewall blocking

Fix steps:
1. Check public IP / NAT Gateway
2. Verify effective routes
3. Check NSG (outbound 0.0.0.0/0)
4. Check Azure Firewall/NVA
5. Test with: Test-NetConnection 8.8.8.8

6 NSG blocks traffic

Symptoms:
- Cannot connect to VM / service
- Connection timeout

Troubleshoot:
1. Check NSG rules (priority, allow/deny)
2. Check effective NSG (NIC + Subnet)
3. Verify source/destination, port, protocol
4. Check NSG flow logs (Log Analytics)
5. Test from another subnet

Example:
Allow HTTPS (443) from 10.0.0.0/16 to a web server.

Interview trap:
Remember both inbound and outbound rules. Deny rules with higher priority override allow rules.

7 UDR misconfigured

Symptoms:
- Traffic not reaching expected destination
- Internet not working
- Going to wrong next hop

Troubleshoot:
1. Check effective routes (NIC/Subnet)
2. Look for 0.0.0.0/0 or specific prefixes
3. Verify next hop type (Virtual appliance, Gateway, Internet)
4. Confirm return route (asymmetric)

Example:
UDR 0.0.0.0/0 → NVA (firewall). If NVA down, internet breaks.

Interview trap:
Forgetting return routes (asymmetric routing) or using wrong next hop type.

8 Private Endpoint inaccessible

Symptoms:
- Cannot reach service via private endpoint
- Name resolves but connection fails

Troubleshoot:
1. Check Private DNS zone and VNet link
2. Verify NIC and subnet (no NSG block)
3. Check service firewall (allow VNet)
4. Test name resolution (nslookup)
5. Validate routing (no forced tunneling to on-prem)

Example:
Access Azure Storage via private endpoint using privatelink.blob.core.windows.net.

Interview trap:
DNS is the most common issue. Also check if UDR is sending traffic to on-prem.

9 DNS failures

Symptoms:
- Cannot resolve name (NXDOMAIN)
- Resolves to wrong IP

Troubleshoot:
1. Test with nslookup or dig
2. Check DNS server settings (VNet / on-prem)
3. For private endpoints, ensure private DNS zone link
4. Check conditional forwarders (hybrid DNS)
5. Flush DNS cache (ipconfig /flushdns)

Example:
VM cannot resolve myapp.privatelink.database.windows.net.
→ Link the private DNS zone to the VNet.

Interview trap:
DNS can resolve but still not connect (network/NSG/firewall issue).

10 VNet peering broken

Symptoms: Cannot communicate between peered VNets

Troubleshoot:
1. Verify peering state (Connected)
2. Check Allow VNet Access (both sides)
3. Check Allow Forwarded Traffic (if using NVAs)
4. Verify routes and NSGs
5. Ensure addressing doesn't overlap
6. Test with network watcher (test-connectivity)

Example:
Peering betweene hub and spoke not working → enable Allow VNet Access and check NSG rules.

Interview trap:
VNet peering is not transitive. Overlapping address spaces are not allowed.

Good Networking makes Applications Happy!

Keep Learning
Keep Building

- [x] Understand
- [x] Practice
- [x] Troubleshoot
- [x] Be Interview Ready!

Azure Load Balancing

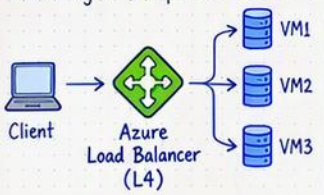
Distribute Traffic • Improve Availability • Scale with Confidence

"Right service for the right layer makes your application resilient!"

Key Goals:
- ☑ High Availability
- ☑ Scalability
- ☑ Better Performance
- ☑ Security
- ☑ Global Reach

1 Azure Load Balancer (Layer 4)

What it is:
Layer 4 (TCP/UDP) load balancer that distributes traffic to backend VMs using IP and port.

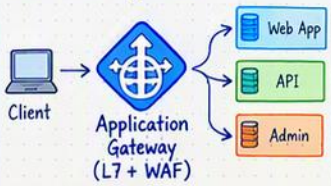


Key features:
- Handles TCP/UDP traffic
- Used for non-HTTP workloads
- Supports HA ports, NAT, internal & external load balancing
- Health probes (TCP/HTTP)

Example:
Distribute traffic to web servers on port 80 or to SQL servers on port 1433.

2 Application Gateway (Layer 7)

What it is:
Layer 7 (HTTP/HTTPS) load balancer with advanced routing and built-in WAF (Web Application Firewall).

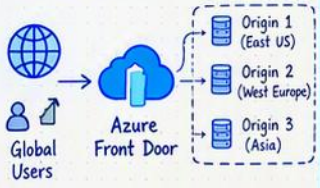


Key features:
- HTTP/HTTPS, path-based routing
- Host-based routing (multiple sites)
- SSL termination
- WAF for application security
- Autoscaling, session affinity

Example:
Route contoso.com/api to API servers and contoso.com to web servers with WAF enabled.

3 Azure Front Door (Global)

What it is:
Global Layer 7 load balancing and acceleration service for HTTP/HTTPS with CDN, WAF and traffic optimization.

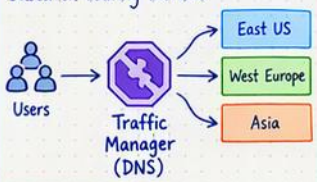


Key features:
- Global load balancing
- Anycast, CDN (caching)
- WAF and DDoS protection
- Traffic routing (latency, health, geo)
- Supports multi-region architectures

Example:
Route users to the nearest healthy region and cache static content using Front Door + CDN.

4 Traffic Manager (DNS)

What it is:
DNS-based load balancing that directs users to the best endpoint based on routing method.



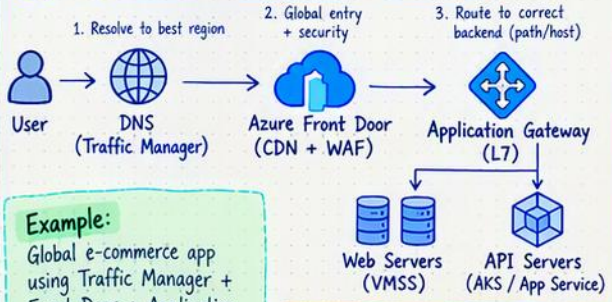
Routing methods:
- Priority (failover)
- Weighted (traffic split)
- Performance (lowest latency)
- Geographic (based on location)
- Multivalue (return multiple IPs)

Example:
Send 80% traffic to East US and 20% to West Europe using weighted routing.

5 Comparison

Feature	Load Balancer (L4)	Application Gateway (L7)	Front Door (Global)	Traffic Manager (DNS)
OSI Layer	Layer 4	Layer 7	Layer 7	DNS (Layer 7)
Protocol	TCP / UDP	HTTP / HTTPS	HTTP / HTTPS	DNS
Main Use Case	Distribute traffic to VMs	Advanced HTTP routing + WAF	Global load balancing + CDN + WAF	DNS-based routing across regions
Routing	IP + Port	Path / Host / URL	Latency / Health / Geo	Priority / Weighted / Performance / Geo
WAF	No	Yes	Yes	No
SSL Termination	No (pass-through)	Yes	Yes	No
Scope	Regional	Regional	Global	Global
Typical Backends	VMs, VMSS, ILB	VMs, App Service, AKS	Multi-region apps, web apps, storage	Any public endpoint

6 Request Flow (Example Architecture)



Example:
Global e-commerce app using Traffic Manager + Front Door + Application Gateway + VMSS/AKS.

Note:
You may not always need all services. Choose based on requirements.

7 Common Issues & Troubleshooting

502 Bad Gateway

Meaning: Gateway received an invalid response from backend.

Common causes:
- Backend server down
- App not responding
- Wrong backend port
- Misconfigured health probe

Troubleshoot:
1. Check backend health
2. Verify app logs
3. Validate probe settings
4. Test direct access to backend

503 Service Unavailable

Meaning: Service is temporarily unavailable.

Common causes:
- No healthy backends
- All instances down
- Maintenance / draining
- Auto scale delay

Troubleshoot:
1. Check health probe status
2. Verify backend pool
3. Check autoscaling/VMSS
4. Review maintenance updates

Health Probe Failure

Meaning: Load balancer marks backend as unhealthy.

Common causes:
- Wrong probe path/port
- App slow to respond
- Firewall blocking probe IPs
- Probe expecting 200 but getting 301/302/404

Troubleshoot:
1. Validate probe configuration
2. Test probe URL/port manually
3. Check NSG/Firewall rules
4. Review app health endpoint

SSL Handshake Issue

Meaning: SSL/TLS negotiation fails between client and gateway.

Common causes:
- Invalid/expired certificate
- TLS version/cipher mismatch
- SNI not configured
- Incomplete certificate chain

Troubleshoot:
1. Check certificate validity
2. Verify supported TLS version
3. Check SNI/host name
4. Use SSL test tools (e.g. ssllabs)

Backend Unhealthy

Meaning: All or some backends are marked unhealthy.

Common causes:
- Application crash
- High CPU / memory
- Network/security rules
- Wrong health probe settings

Troubleshoot:
1. Check backend instance status
2. Review metrics and logs
3. Validate NSG/UDR rules
4. Ensure app is listening on expected port

8 Key Takeaways

- ☑ Understand the right load balancing service for your use case.
- ☑ Use health probes and monitoring to detect failures early.
- ☑ Combine services (e.g., Front Door + App Gateway) for global, secure, and scalable solutions.
- ☑ Always test failure scenarios (region outage, backend down).
- ☑ Keep security in mind – use WAF, DDoS protection, and TLS best practices.
- ☑ There is no one-size-fits-all – design based on application requirements.

Pro Tip:
Start simple, then evolve. You can begin with a Load Balancer and move to Application Gateway or add Front Door as your needs grow.

"High availability is not a feature, it's a design choice!"

Azure Compute

Run, Scale and Manage Applications in the Cloud

"Compute is the engine that turns ideas into impact!"

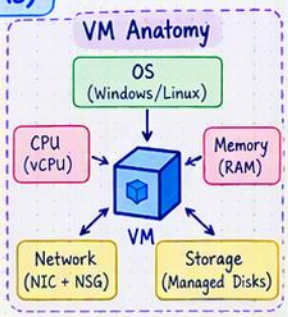
- [x] Learn
- [x] Practise
- [x] Build
- [x] Get Certified
- [x] You can do it!

1 Virtual Machines (VMs)

What it is: Resizable compute instances in Azure. You choose the OS, size, storage, networking and manage the VM.

Use cases:

- Lift and shift applications
- Custom configurations
- Dev/Test environments
- Applications needing full control



Example:
Create a Windows VM called web-vm in East US with a Standard B2s size.

```
az vm create -g rg-demo -n web-vm --image Win2022
  --size Standard_B2s --location eastus
```

Remember:

- VMs are IaaS – you manage the OS and updates.
- Use Managed Disks for better reliability and SLA.

2 VM Scale Sets (VMSS)

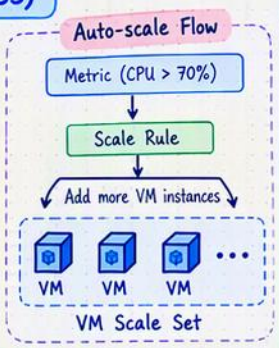
What it is: Automatically scales a group of identical VMs based on demand.

Benefits:

- Auto-scale (increase/decrease instances)
- Load balancing across instances
- Self-healing (recreates failed VMs)
- Cost effective for scalable applications

Use cases:

- Web applications, APIs
- Batch processing, big data
- Stateless workloads



Example: Scale out when CPU > 70% (adds 2 instances)

```
az vmss create -g rg-demo -n web-vmss --image Ubuntu2204
  --instance-count 2 --upgrade-policy-mode automatic
```

Remember:

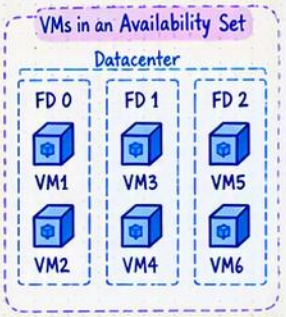
- VMSS is for identical VMs.
- Supports auto-scaling, rolling upgrades, self-healing.

3 Availability Sets

What it is: Logical grouping of VMs within a datacenter to protect against hardware failures.

Key concepts:

- Fault Domains (FD) – separate racks (power/network)
- Update Domains (UD) – groups for planned maintenance
- SLA: 99.95% (when VMs across multiple FDs and UDs)



Example:
Create an availability set with 3 fault domains and 5 update domains.

```
az vm availability-set create -g rg-demo -n web-as \
  --platform-fault-domain-count 3 --platform-update-domain-count 5
```

Remember:

- Protects against hardware and planned maintenance failures within a datacenter.

4 Availability Zones (AZ)

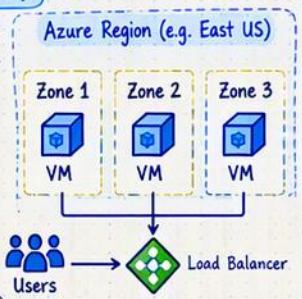
What it is: Physically separate datacenters within a region. Each zone has independent power, cooling and networking.

Benefits:

- Higher resilience to datacenter failures
- SLA: 99.99% (for VMs across zones)
- Low latency between zones (< 2ms)

Use cases:

- Mission-critical applications
- Zone-redundant services (e.g., Azure SQL)
- Applications requiring higher availability



Example:
Create a VM in a specific zone (Zone 1).

```
az vm create -g rg-demo -n db-vm --zone 1 \
  --image Ubuntu2204 --size Standard_B2s
```

Remember:

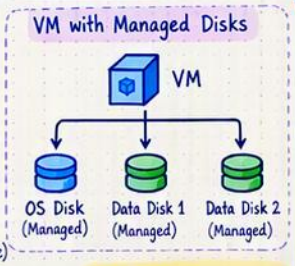
- Zones protect against datacenter level failures within a region.

5 Managed Disks

What it is: Persistent, durable block storage for Azure VMs. Managed by Azure (no need to manage storage accounts).

Disk types:

- Standard HDD – cost effective
- Standard SSD – general purpose
- Premium SSD – high performance, low latency
- Ultra Disk – very high performance (I/O intensive)



Example: Attach a new data disk to a VM.

```
az disk create -g rg-demo -n data-disk --size-gb 128 --sku Standard_LRS
az vm disk attach -g rg-demo -n web-vm --disk data-disk
```

Remember:

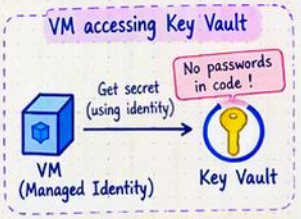
- Managed Disks offer built-in redundancy and encryption.
- Choose the right disk type based on performance needs.

6 Managed Identity

What it is: Provides an Azure AD identity to a VM/VMSS so it can access Azure resources securely without storing credentials in code or configuration.

Types:

- System-assigned – tied to the resource
- User-assigned – can be reused across multiple resources



Example: Give VM access to Key Vault secrets.

```
az role assignment create --assignee <vm-identity-id> \
  --role "Key Vault Secrets User" --scope /subscriptions/<sub-id>/
  resourceGroups/rg-demo/providers/Microsoft.KeyVault/vaults/kv-demo
```

Remember:

- More secure than using service principals or storing keys.
- Follow least privilege access.

7 Common Interview Questions

Q1. When to use Availability Sets vs Availability Zones?
Q2. What is the difference between a VM and a VM Scale Set?
Q3. Which disk type should you choose for a high performance database?
Q4. How does Managed Identity improve security?
Q5. What SLA can you achieve with VMs across multiple Availability Zones?
Q6. Can you move a VM from one zone to another?

Key Takeaways

- [x] Choose the right compute option for the workload.
- [x] Use WMSS for automatic scaling.
- [x] Use Availability Sets or Zones for high availability.
- [x] Use Managed Disks for reliable storage.
- [x] Use Managed Identity to access Azure resources securely.
- [x] Consider cost, performance and SLA requirements.
- [x] Follow security best practices (NSG, RBAC, least privilege).

"Right Compute + Right Design = Scalable, Resilient and Cost-Effective Applications"

Azure Storage

Scalable • Secure • Highly Available • Cost Effective

Store Anything
Build Everything on Azure

"Right data
Right tier
Right access
Right cost !

1 Azure Storage Types

Different storage for different needs !

Blob Storage

Blob

- Store unstructured data (text, images, video, backups)
- Supports block, append and page blobs
- Highly scalable and cost effective

Example: Store product images for an e-commerce website.

Azure Files

- Managed file shares in the cloud (SMB/NFS)
- Mount from Windows, Linux or macOS
- Good for lift-and-shift applications

Example: Share company files between VMs or on-prem via VPN.

Azure Queues

- Store messages for asynchronous communication
- Decouple applications
- Gimple, reliable and low cost

Example: Web app adds a message to queue for background image processing.

Azure Tables

- NoSQL key-value store for structured data
- Schema-less and highly scalable
- Great for IoT, user telemetry, metadata

Example: Store device telemetry data from IoT sensors.

2 Azure Data Lake Storage Gen2 (ADLS Gen2)

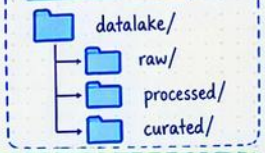
What it is: A high-performance storage for big data analytics with a hierarchical namespace (like a file system).

- Built on top of Blob Storage
- Supports HDFS compatible APIs
- Optimized for big data and analytics (Azure Databricks, Synapse, HDInsight)
- Fine-grained access control with Azure RBAC and ACLs

Use cases:

- Data lakes
- Big data analytics
- Machine learning datasets
- Log and event data

Hierarchical Namespace



Example: Store sales data in raw/ folder, process it using Databricks and save results in curated/ folder.

3 Storage Redundancy Options

Keeps your data safe from failures !

LRS (Local Redundant Storage)

- 3 copies in one datacenter
- Protects against hardware failures
- Lowest cost

3 copies (1 datacenter)

ZRS (Zone Redundant Storage)

- 3 copies across 3 availability zones
- Protects against datacenter failures
- Higher cost

3 copies (3 zones)

GRS (Geo-Redundant Storage)

async

- 3 copies in primary region + 3 copies in paired region
- Protects against regional disasters
- Asynchronous replication

Primary Region | Paired Region

6 copies (async)

GZRS (Geo-Zone-Redundant Storage)

async

- 3 copies across zones in primary + 3 copies in paired region
- Highest resilience
- Protects against zone and regional failures

Primary (3 zones) | Paired (3 zones)

6 copies (async)

4 SAS Tokens (Shared Access Signatures)

What it is: A time-limited token that provides delegated access to storage resources without sharing account keys.

```
https://<account>.blob.core.windows.net/<container>/<blob>
?sv=2023-08-03&sr=b&sp=r&se=2024-12-31T23:59:59Z&sig=...
```

Permissions (sp):

r = read	w = write	d = delete
l = list	a = add	c = create
u = update	p = process	

Use cases:

- Securely share files with external users
- Provide temporary access
- Grant limited permissions

Important:

- Set a short expiry time
- Use least privilege permissions
- Prefer User Delegation SAS (AAD) over Account Key SAS
- Rotate/revoke when no longer needed

5 Lifecycle Management

Move data to cheaper tiers automatically !

Hot — Frequently accessed data (Highest cost)
→ After 30 days
Cool — Infrequently accessed data (Lower cost)
→ After 180 days
Archive — Rarely accessed data (Lowest cost)
→ After retention period
Delete — Permanently delete data

Example Policy:
Move blobs to Cool after 30 days, to Archive after 180 days and delete after 2 years.

Supported for:

- Blob Storage (block blobs)
- ADLS Gen2

6 Storage Decision Guide

Which storage should I use?

Scenario / Use case	Recommended storage
Store files, images, videos, backups	Blob Storage
Shared file system for VMs / on-prem	Azure Files
Asynchronous messaging between apps	Azure Queues
Structured NoSQL data (Key-value)	Azure Tables
Big data, analytics, data lake	ADLS Gen2
Need lowest cost and infrequent access	Blob (Cool / Archive)
Need cross-region disaster recovery	GRS / GZRS
Share data with external users (temp access)	SAS Token

7 Key Takeaways

- [x] Choose the right storage type for the workload.
- [x] Use appropriate redundancy based on RPO/RTO.
- [x] Use lifecycle management to optimize cost.
- [x] Secure access using RBAC, SAS and/or private endpoints.
- [x] For analytics and big data, use ADLS Gen2.
- [x] Monitor usage and costs.
- [x] Always follow least privilege and defense in depth.

Store smart
Scale fast
Save cost
Stay secure

That's Azure Storage !

8 Common Interview Questions / Traps

Q: What's the difference between Blob and ADLS Gen2?
A: ADLS Gen2 has hierarchical namespace and is optimized for big data analytics. Blob is for general object storage.

Q: Which redundancy should I choose?
A: Depends on your RPO/RTO. LRS for low cost, ZRS for zone protection, GRS/GZRS for regional disaster recovery.

Q: Can we use SAS instead of RBAC?
A: SAS is for delegated, time-limited access. RBAC is for managing access using Azure AD identities. Both serve different purposes.

Q: Is Archive storage immediately retrievable?
A: No. Archive requires a rehydration process which can take hours and may incur additional charges.

"Data is the foundation of modern applications"

Keep Learning →

People
Apps
Services
Devices

Azure Identity and Security

Secure Identities → Protect Resources → Enable Zero Trust

Security is a feature, not an add-on!

1 Microsoft Entra ID (Azure AD)

What it is:
Cloud identity and access management service for users, groups, apps and devices.

Microsoft Entra ID

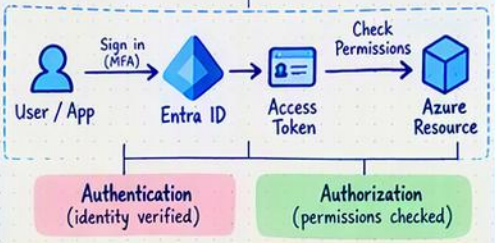
- Provides authentication, SSO and conditional access
- Supports MFA, identity protection
- Integrates with Microsoft 365, Azure and many SaaS apps

Example:
- Users sign in to Azure portal with Entra ID.
- Use Conditional Access to require MFA.

Interview trap:
Entra ID is not the same as an Azure subscription. One tenant can have multiple subscriptions.

2 Authentication vs Authorization

Authentication	Authorization
Who are you?	What can you do?
• Validates identity (user, app, device)	• Checks permissions (RBAC, policies)
• Uses credentials, MFA, tokens	• Determines access to resources



Remember: You must be authenticated before you can be authorized!

3 Azure RBAC

What it is:
Role Based Access Control to manage access to Azure resources.

- Assign roles at different scopes (MG, Subscription, RG, Resource)
- Built-in and custom roles
- Follows least privilege principle

Principal + Role + Scope = Access

Example:
Assign "Virtual Machine Contributor" role to a user at resource group level.

Interview trap:
RBAC controls "what you can do", not "how you sign in" (that's Entra ID).

4 Managed Identity

What it is:
Identity for Azure services to access other Azure resources without storing credentials in code.

Managed Identity

- System-assigned (tied to resource)
- User-assigned (reusable across resources)
- Uses Entra ID under the hood

Example:
An Azure VM uses a managed identity to access Key Vault and read secrets.

Interview trap:
A managed identity cannot be used outside Azure.

5 Service Principal

What it is:
Identity for applications, services or automation (non-human).

Service Principal

- Created in Entra ID
- Uses client ID + client secret or certificate
- Commonly used by CI/CD, scripts and external apps

Example:
Azure DevOps uses a service principal to deploy resources using Azure CLI.

Interview trap:
A service principal is not a managed identity. It requires secret or certificate (except federated).

6 Azure Key Vault

What it is:
Securely stores secrets, keys, certificates and other sensitive data.

Azure Key Vault

- Integrates with Entra ID (RBAC or Access Policies)
- Supports soft delete and purge protection
- Used for secrets, encryption keys, TLS certs

Example:
Store database connection string and let an App Service (managed identity) read it.

Interview trap:
Do not store non-sensitive data in Key Vault. Always enable soft delete and purge protection.

7 Azure Policy

What it is:
Enforces rules and compliance across your Azure environment.

Azure Policy

- Define policies (e.g., allowed regions, required tags, VM SKUs)
- Assign at MG, subscription or RG
- Can audit, deny or modify resources

Example:
Require all resources to have a "Environment" tag.

Interview trap:
Policy does not replace RBAC. It enforces rules, while RBAC controls access.

8 Microsoft Defender for Cloud

What it is:
Unified security management and threat protection for Azure (and multi-cloud).

Defender for Cloud

- Continuous security assessment
- Threat detection & recommendations
- Integrates with Sentinel, Log Analytics
- Helps with compliance (e.g., CIS, ISO, NIST)

Example:
Get recommendations for unpatched VMs or open network ports.

Interview trap:
Defender for Cloud is not an antivirus for VMs. It provides security posture and threat detection.

9 Common Issues & Troubleshooting

Key Vault access denied
- [x] Check identity (user/SP/managed identity)
- [x] Verify RBAC or access policy on Key Vault
- [x] Ensure correct subscription and tenant
- [x] Check firewall / private endpoint settings
- [x] Review diagnostic logs

Expired service principal
- [ ] Error: AADSTS7000222 (expired credential)
- [ ] Check app registration -> Certificates & secrets
- [ ] Renew client secret or certificate
- [ ] Update in your pipeline / application
- [x] Consider using certificate or federated credentials

RBAC issue (no access)
- [ ] Verify your user/service principal
- [x] Check role assignment and scope
- [x] Use "az role assignment list" to confirm
- [ ] Consider propagation delay (few minutes)
- [ ] Check for conflicting deny assignments (Azure Policy)

Key Takeaways

- [x] Use least privilege
- [x] Prefer managed identity over secrets
- [x] Enable MFA and Conditional Access
- [x] Use Key Vault for sensitive data
- [x] Monitor with Defender for Cloud
- [x] Follow Zero Trust principles

"Identify. Protect. Detect. Respond."
A more secure tomorrow on Azure!



Same Concepts Bigger Opportunities in the Cloud!

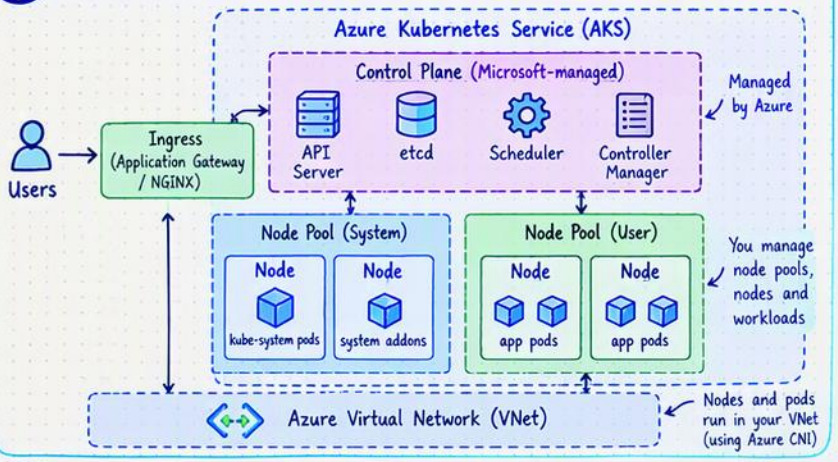
Azure Kubernetes Service - AKS

Managed Kubernetes on Azure → Deploy · Scale · Secure · Operate

Less Ops work More time to build !

AKS is a managed Kubernetes service by Azure. It handles control plane management (API server, etcd, scheduler) while you manage the worker nodes, node pools and your applications.

1 AKS Architecture (Control Plane + Node Pools)



2 Node Pools

What it is: A node pool is a group of identical worker nodes (VMs) in an AKS cluster.

Why use it?

- Different workloads (system vs apps)
- Different VM sizes (CPU/Memory/GPU)
- Isolate workloads (e.g., sensitive or batch)
- Enable independent scaling and upgrades

Multiple node pools = Flexibility

Example:

- System node pool: runs essential components (ingress, dns, metrics-server)
- User node pool: runs your application workloads

Interview tip:

- Have at least 2 node pools (system + user) in production.

3 Azure CNI (Networking)

What it is: Azure CNI gives each pod a real IP address from your VNet. Pods can communicate directly with other Azure resources.



Example: Pod gets IP 10.244.0.5 from VNet subnet and can access Azure SQL, Key Vault or on-prem resources.

Interview tip: Know the difference: Azure CNI vs kubenet. Azure CNI is preferred for production.

4 ACR Integration (Container Registry)

What it is: Azure Container Registry (ACR) stores your container images. AKS pulls images from ACR to run your pods.



Example: Build a Docker image, push to ACR, then create a Kubernetes deployment that pulls the image from ACR.

Useful Commands:

```
az acr create -n myacr -g rg -l eastus
az acr login -n myacr
docker build -t myacr.azurecr.io/myapp:v1 .
docker push myacr.azurecr.io/myapp:v1
```

Use Managed Identity for secure access (no secrets!)

5 Azure Monitor for AKS

What it is: Integrates AKS with Azure Monitor (Log Analytics) for metrics, logs and insights.

What you get:

- Container logs (stdout, stderr)
- Cluster performance metrics (CPU, memory, nodes)
- Kubernetes events and audit logs
- Integration with Azure Managed Prometheus & Grafana
- Alerts and dashboards

Logs
Metrics
Alerts
Dashboards
Insights

Example: View pod logs and node metrics in Azure Portal → Monitor → Containers or use kubectl commands.

Interview tip: Know how to troubleshoot a failing pod using logs and events in Azure Monitor.

6 Cluster Autoscaler (Scale Nodes)

What it is: Automatically adjusts the number of nodes in a node pool based on pending pods and resource usage.

Scale Out (add nodes)

How it works?

1. New pods are pending (no resources).
2. Cluster Autoscaler adds nodes (up to max).
3. When load reduces, it removes idle nodes (down to min).



Scale In (remove nodes)

Example: Node pool min = 1, max = 5. When more pods are scheduled, AKS adds nodes automatically.

Interview tip: Cluster Autoscaler scales nodes, not pods. For pod scaling, use Horizontal Pod Autoscaler (HPA).

HPA vs Cluster Autoscaler

Different level Different purpose Both needed!

Feature	HPA (Horizontal Pod Autoscaler)	Cluster Autoscaler
Scales	Pods (application level)	Nodes (infrastructure level)
Based on	CPU/Memory/custom metrics	Pending pods / node resource usage
Managed by	Kubernetes (you configure)	AKS (enabled per node pool)
Example	Increase pod replicas from 2 → 10	Add nodes from 2 → 5

Key Takeaways

- ✅ AKS is a managed Kubernetes service (Azure manages control plane).
- ✅ Use multiple node pools for isolation and flexibility.
- ✅ Use Azure CNI for VNet integration and real pod IPs.
- ✅ Store container images in ACR and use managed identity for access.
- ✅ Use Azure Monitor for logs, metrics and alerting.
- ✅ Use HPA to scale pods and Cluster Autoscaler to scale nodes.
- ✅ Design for security, monitoring and cost optimization.

Plan Deploy Scale Monitor Secure Succeed !

Kubernetes on Azure = Happy DevOps!

Containers today → Scalable applications tomorrow !

Azure Monitoring

Observe → Understand → Act → Keep Improving

"You can't improve what you don't measure!"

Monitoring is not just logs, it's peace of mind in production!

Collect telemetry, gain insights, detect issues and take action to keep your applications healthy.

1 Azure Monitor

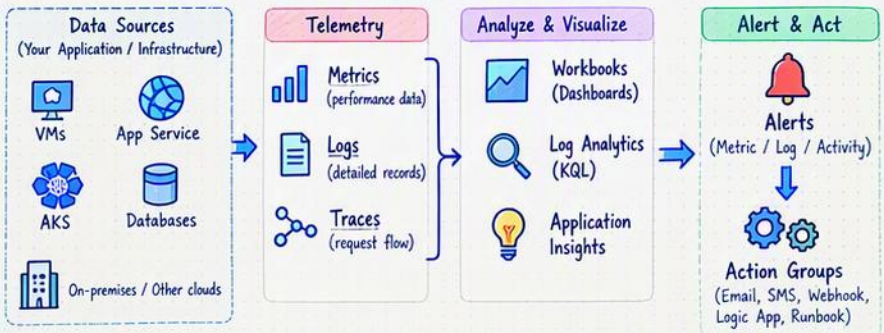
What it is? Azure Monitor is a unified monitoring platform for metrics, logs, traces and other telemetry from Azure resources, on-premises and other clouds.

Key features:

- Collects metrics, logs, traces, activity logs
- Built-in dashboards and workbooks
- Supports alerting and automation
- Integrates with Log Analytics, Application Insights, and more

Example:
Monitor a VM, App Service and AKS cluster in a single pane using Azure Monitor.

Telemetry Pipeline in Azure



2 Log Analytics and KQL

What it is? Log Analytics is the data store for logs in Azure Monitor. Kusto Query Language (KQL) is used to query and analyze log data.

Why use it?

- Query large volumes of log data quickly
- Troubleshoot issues
- Create alerts based on log queries
- Visualize data in workbooks

KQL = SQL for big data logs!

Example KQL query:

```
requests
| where timestamp > ago(1h)
| where success == false
| project timestamp, name, resultCode
| order by timestamp desc
```

3 Application Insights

What it is? Application Insights is an application performance monitoring (APM) service that collects telemetry from your application (code), shows request flow, dependencies, exceptions and performance.

Key capabilities:

- Automatic instrumentation (SDK)
- Distributed tracing (end-to-end view)
- Request & dependency tracking
- Exception and failure analysis
- User and session insights
- Integration with Log Analytics

Example:
Use Application Insights in a web app to find slow pages and failed dependencies (e.g., SQL, HTTP, external APIs).

4 Alerts and Action Groups

What it is? Alerts notify you when a condition is met (e.g., high CPU, error rate) and Action Groups take the required action (email, SMS, webhook, Logic App, runbook, ITSM, etc.).

Alert → Notify → Take Action → Resolve Faster!

Types of alerts:

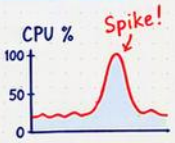
- Metric alerts – based on metrics
- Log alerts – based on KQL queries
- Activity log alerts – based on Azure activity
- Smart alerts – ML based (Application Insights)

Example:
If CPU > 80% for 5 minutes, send an email and trigger an Azure Automation runbook to scale out.

5 Troubleshooting: CPU Spike

Symptoms:

- High CPU on VM / App Service
- Application slow or unresponsive



Troubleshooting steps:

1. Check metrics in Azure Monitor (CPU, Memory)
2. Identify time window and correlate with logs
3. Use KQL to find top processes or requests
4. Check dependencies (DB, external APIs)
5. Scale up or scale out if needed

Correlate metrics + logs to find the real cause!

Example (KQL):

```
Perf
| where CounterName == "% Processor Time"
| summarize avg(CounterValue) by bin(TimeGenerated, 5m)
```

6 Troubleshooting: Missing Logs

Symptoms:

- Logs not appearing in Log Analytics
- Partial logs or delayed ingestion

No logs?

Troubleshooting steps:

1. Check if Diagnostic Settings are enabled
2. Verify the correct Log Analytics workspace
3. Check data source / agent / extension status
4. Look for ingestion errors (Ingestion logs)
5. Verify permissions (e.g., RBAC)
6. Allow some time (can take a few minutes)

Check Diagnostic Settings first!

Common fixes:

- Enable diagnostic settings for the resource
- Check workspace ID and region
- Ensure agent/extension is healthy
- Check network/firewall if using private link

7 Troubleshooting: Alert Not Firing

Symptoms:

- Condition met, but no alert triggered

Why didn't my alert fire?

Troubleshooting steps:

1. Check alert rule configuration
   - Correct metric / log query
   - Threshold and operator
   - Evaluation frequency and time window
2. Check scope (right resource/region)
3. Verify the rule is Enabled
4. Check Action Group configuration
5. Look at the Alert rule history
6. For log alerts, test the KQL query manually

Test the query and check the alert history!

8 Request and Dependency Trace (Application Insights)



Helps you:

- See end-to-end request flow
- Identify slow dependencies
- Find failures and exceptions
- Improve performance

Key Takeaways

- Use Azure Monitor for unified observability.
- Leverage Log Analytics and KQL for deep analysis.
- Use Application Insights for application performance.
- Configure meaningful alerts and automate actions.
- Correlate metrics, logs and traces to troubleshoot.
- Monitoring helps you detect issues early and keep users happy!

Monitor Today for a Better Tomorrow!

"Good monitoring turns problems into insights, and insights into better solutions."

High Availability and Disaster Recovery

Keep your applications running, no matter what!

"Plan for Failures, Build for Resilience!"

- High Availability
- Business Continuity
- Minimal Downtime

Why it matters?

- Minimize downtime and data loss
- Keep business running during failures
- Meet compliance and SLA requirements
- Be prepared for planned and unplanned outages

"Failure is not an option, It's a scenario to plan for."

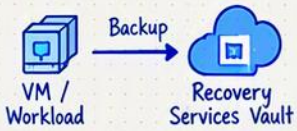
Common failure scenarios:

- Datacenter outage
- Region-wide disaster (natural disaster)
- Application failure
- Human error (accidental deletion)
- Security incidents (ransomware)

1 Azure Backup

What it is: A cloud-based backup service to protect your data and workloads in Azure, on-premises and other clouds.

- Supports VMs, databases, files, AKS, etc.
- Stores backup data in a Recovery Services vault
- Supports scheduled and on-demand backups
- Retention policies and long-term storage (vault, archive)



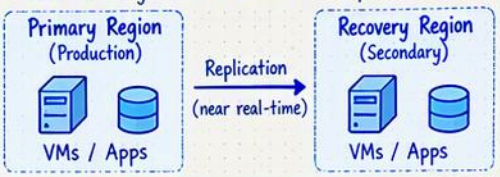
- Backup policy
- Reterption
- Encryption
- Restore when needed

Example: Daily backup of an Azure VM with 30-day retention. Restore to the same or different region.

2 Azure Site Recovery (ASR)

What it is: A disaster recovery service that replicates VMs, physical servers and other workloads to a secondary region and helps failover during outages.

- Replicates VMs at near real-time
- Orchestrates failover and failback
- Supports Azure to Azure, on-premises to Azure
- Uses Recovery Plans to automate the process



Example: Replicate on-premises VMs to Azure using ASR. During a disaster, failover to Azure with minimal downtime.

Key Benefits:

- Minimal downtime
- Automated failover
- Supports multi-region
- Integrated with Azure Backup

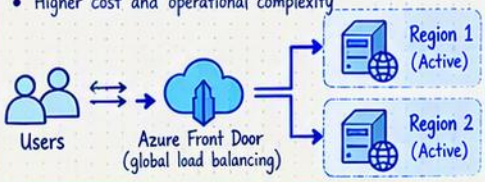
Remember: ASR is for disaster recovery, not a backup replacement.

Interview Trap: ASR replicates disks in near real-time. It does NOT keep historical versions (like Backup does).

3 Active – Active (Multi-Region)

What it is: Application runs in multiple regions at the same time, actively serving traffic.

- Provides high availability and better user experience
- Uses DNS (Traffic Manager / Azure Front Door)
- Requires data replication and conflict handling
- Higher cost and operational complexity

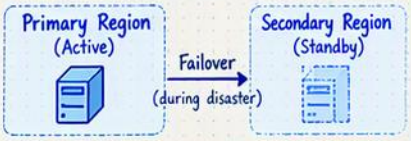


Example: Global web application using Azure Front Door with instances in East US and West Europe, both active.

4 Active – Passive (Primary-Secondary)

What it is: Application runs in one region (primary) and a standby environment in another region (passive).

- Lower cost than active-active
- Standby environment is ready for failover
- Can be warm (scaled down) or cold (minimal resources)
- Uses ASR, geo-replication, or database replication



Example: Main application in Central India, standby environment in South India. In case of outage, failover to secondary region.

5 RPO and RTO

RPO (Recovery Point Objective)

- Maximum acceptable data loss.
- Measured in time (e.g. 5 mins, 1 hour)
- Depends on backup/replication frequency.

RTO (Recovery Time Objective)

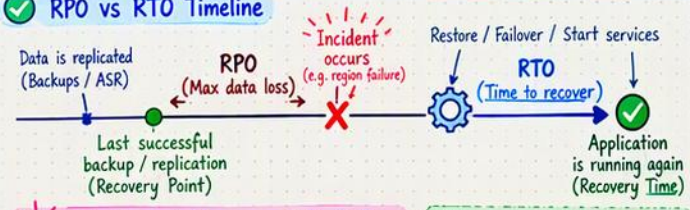
- Maximum acceptable downtime.
- Time to restore and resume service.
- Depends on failover process and architecture.

Lower RPO and RTO = higher cost (more replication, resources, automation).

Typical values (example):

Mission critical:	RPO: seconds	RTO: minutes
Business critical:	RPO: minutes	RTO: hours
Non critical:	RPO: hours	RTO: hours/days

RPO vs RTO Timeline



RPO = How much data can you afford to lose?
RTO = How quickly do you need to be back online?

RPO is measured BEFORE the incident.
RTO is measured AFTER the incident.

Comparison Table

Feature	Active – Active	Active – Passive
Regions used	Multiple (simultaneous)	Primary + Secondary
Traffic	Live in all regions	Only primary (standby idle)
Cost	Higher	Lower
Complexity	Higher	Lower
Use case	Global apps, zero downtime	Disaster recovery
RPO / RTO	Near zero / very low	Depends on replication and failover (minutes to hours)

Recovery Checklist

- [x] Define RPO and RTO based on business needs
- [x] Choose the right solution (Backup, ASR, geo-replication, etc.)
- [x] Test backups and restore process regularly
- [x] Use Recovery Plans for automated failover (ASR)
- [x] Validate application functionality after failover
- [x] Document and run DR drills
- [x] Monitor and improve

Interview Questions

- What is the difference between Azure Backup and ASR?
- When would you choose active-active vs active-passive?
- How do you test your DR plan?
- What factors influence RPO and RTO?
- Can ASR be used for databases?
- How does Azure Front Door help in HA/DR?

Common Interview Traps

- ASR is not a backup solution.
- Backup does not provide real-time failover.
- Active-active is not always required (consider cost).
- RPO is not the same as RTO.
- Not testing DR (always mention periodic testing).
- Assuming all Azure services support zone/region redundancy (check service limits).
- Forgetting about data consistency and replication lag in active-active setups.

Resilient Applications Happier Customers!

Plan • Protect • Test • Recover • Repeat

Azure Architecture Design

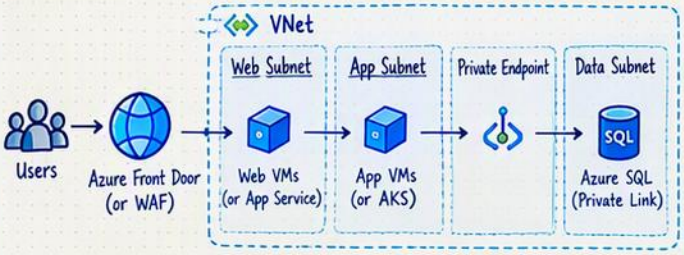
Design Scalable, Secure, Resilient and Cost-Effective Solutions

Good Architecture Turns Requirements into Reliable Solutions!

Key Principles:
- Reliability
- Security
- Scalability
- Cost Optimization
- Operational Excellence
- Well-Architected Framework

1 3-Tier Web Application (Typical Azure Design)

What it is: A common architecture that separates the application into 3 logical tiers – Presentation (Web), Business Logic (App) and Data, deployed in subnets within a VNet.

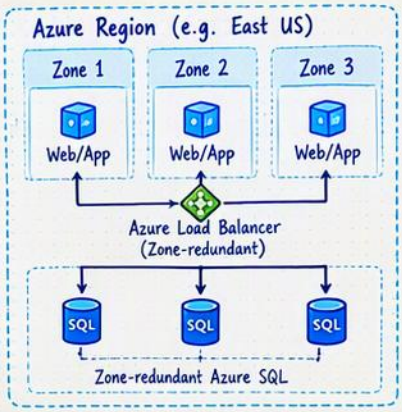


Why use it:
- Separation of concerns
- Improved security (subnets/NSGs)
- Scalable and maintainable
- Can replace VMs with PaaS (App Service/AKS)

Example: Contoso e-commerce app with web tier in App Service, business logic in AKS and Azure SQL using Private Link.

2 Highly Available Platform (Zone-Resilient)

What it is: Designing workloads across multiple Availability Zones within a region to protect against datacenter failures.

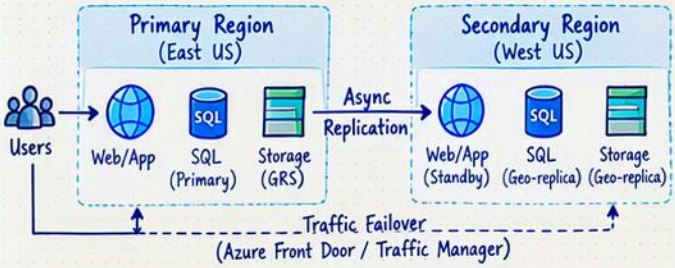


Key Points:
- Deploy across 2+ zones for 99.99% SLA (VMs)
- Use zone-redundant services (LB, App Service, SQL, Storage)
- Automatic failover within the region
- Low latency between zones (< 2ms)

Remember: Zones protect against datacenter failure, not region-wide disasters.

3 Multi-Region Deployment (Disaster Recovery)

What it is: Deploying the application in a primary and secondary region (using region pairs) for business continuity and disaster recovery.

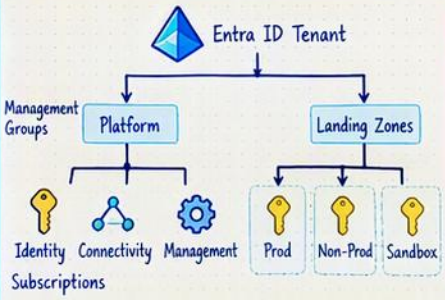


Use cases:
- Protect against region-wide disasters
- Meet compliance / data residency needs
- Zero or minimal downtime with automated failover

Example: Use Azure Front Door for global routing with health probes and automatic failover to secondary region.

4 Secure Enterprise Azure Landing Zone

What it is: A foundational, scalable and secure environment following Microsoft best practices (CAF) for multiple subscriptions and workloads.



Key Components:
- Management groups
- Separate subscriptions (Prod/Non-Prod)
- Hub-and-Spoke networking
- Centralized identity and security
- Policy, RBAC and tagging
- Shared services (monitoring, logging, backup)

Example: Enterprise with a platform subscription (network, security, monitoring) and separate workload subscriptions for each business unit.

5 Security, Monitoring, Backup and Cost Considerations

Security
- Use Entra ID + RBAC
- Apply Azure Policy and resource locks
- Use NSGs, WAF, Private Endpoints
- Enable Defender for Cloud
- Follow Zero Trust principles

Monitoring
- Use Azure Monitor and Log Analytics
- Collect metrics, logs and alerts
- Use Application Insights
- Set up alerting and dashboards
- Monitor cost, health and security

Backup & DR
- Use Azure Backup for VMs and files
- Enable geo-redundant storage (GRS)
- Use point-in-time restore (e.g. Azure SQL)
- Test restore regularly
- Define RPO and RTO

Cost Optimization
- Right-size resources
- Use reserved instances or savings plans
- Use auto-scale
- Remove unused resources
- Tag resources for cost tracking
- Use Cost Management + budgets

6 Key Takeaways & Interview Tips

- [x] Use the right service for the requirement (PaaS where possible).
- [x] Design for HA (zones) and DR (regions).
- [x] Keep the environment secure by design (identity, network, policy).
- [x] Follow the Azure Well-Architected Framework.
- [x] Consider cost from the beginning.
- [x] Diagram and explain the data flow and failure scenarios.
- [x] Mention trade-offs (cost vs performance vs complexity).
- [x] Use real-world examples (e.g. Contoso web app).

Plan Design Implement Operate Improve Repeat!

Architecture is not just about technology, but about making the right trade-offs for your business.

Happy Learning!

Learn Practice Revise Succeed!

Azure Interview Quick Revision

Key Comparisons | CLI Commands | Decision Guide | Final Checklist

Same Concepts Bigger Opportunities You Got This!

1 Rapid Comparisons (Most Asked in Interviews)

Know the differences → Explain with examples → Show use cases

1 NSG vs Azure Firewall

NSG	Azure Firewall
• Layer: L3/L4	• Layer: L3/L4/L7
• Applied to: Subnet/NIC	• Applied to: VNet (central)
• Stateful: Yes	• Stateful: Yes
• Use case: Basic traffic filtering (allow/deny)	• Use case: Advanced threat protection, FQDN, TLS inspection
• Cost: Low	• Cost: Higher

NSG = simple filtering | Firewall = centralized, advanced control

2 Load Balancer vs Application Gateway

Azure Load Balancer	Application Gateway
• Layer: L4 (TCP/UDP)	• Layer: L7 (HTTP/HTTPS)
• Use: Distribute traffic to VMs/VMSS	• Use: Web traffic (routing)
• Supports: Internal & External	• Supports: WAF, SSL offload, path-based routing, host-based
• Use case: High performance, low latency	• Use case: Web apps, microservices
• Features: Health probes, NAT, HA	• Features: WAF, URL rewrite, SSL termination
• Cost: Lower	• Cost: Higher

Load Balancer = L4 & high performance | App Gateway = L7 & web features

3 Front Door vs Traffic Manager

Azure Front Door	Traffic Manager
• Layer: L7 (Application)	• Layer: DNS (L3/L4)
• Scope: Global (edge locations)	• Scope: Global (DNS)
• Use: App acceleration, WAF, SSL offload	• Use: Route users to best region (latency, priority, geo, weighted)
• Routing: Path/host based	• Routing: DNS-based
• Best for: Global web apps	• Best for: Multi-region applications
• Cost: Higher	• Cost: Lower

Front Door = content delivery + security | Traffic Manager = DNS routing

4 VPN vs ExpressRoute

VPN (Site-to-Site)	ExpressRoute
• Connection: Internet (IPsec)	• Connection: Private circuit (service provider)
• Bandwidth: Up to ~1.25 Gbps	• Bandwidth: 50 Mbps – 100 Gbps
• Use case: Small/medium and cost-sensitive	• Use case: Enterprise, critical workloads
• Setup: Quick	• Setup: Longer (provider)
• Reliability: Internet dependent	• Reliability: High (SLA)
• Cost: Lower	• Cost: Higher

VPN = use Internet | ExpressRoute = private, reliable, high bandwidth

5 Private Endpoint vs Service Endpoint

Private Endpoint	Service Endpoint
• Access: Private IP in VNet	• Access: Extends VNet identity
• Traffic: Stays on Microsoft backbone	• Traffic: Microsoft backbone
• Supported: Most PaaS services	• Supported: Selected services (e.g., Storage, SQL, Key Vault)
• Use case: Highest security (no public access)	• Use case: Secure access with VNet identity
• DNS: Uses private DNS zone	• Public access: Still used
• Cost: Higher	• Cost: No extra cost

Private Endpoint = private IP | Service Endpoint = VNet Identity

6 Managed Identity vs Service Principal

Managed Identity	Service Principal
• Created by: Azure (for Azure resources)	• Created by: You (in Entra ID)
• Credential: No secrets to manage	• Credential: Secret or certificate
• Use case: Access Azure resources (e.g., Key Vault)	• Use case: Apps, CI/CD, external tools
• Types: System-assigned, User-assigned	• Lifecycle: Manual management
• Best for: Azure services	• Best for: Non-Azure apps and automation

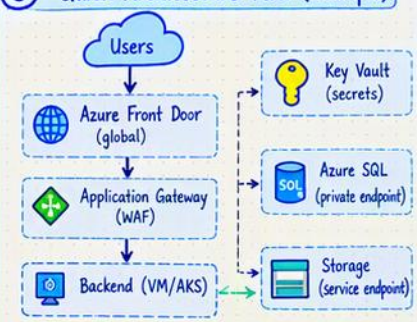
Managed Identity = no secrets | Service Principal = app identity with credentials

7 RBAC vs Azure Policy

RBAC (Role-Based Access Control)	Azure Policy
• Purpose: Manage access (who can do what)	• Purpose: Enforce rules (what can be created)
• Scope: Management group, subscription, RG, resource	• Scope: Management group, subscription, RG
• Examples: Reader, Contributor, Owner	• Examples: Allowed locations, required tags, SKU restrictions
• Controls: Permissions	• Controls: Governance & compliance
• Use case: Access management	• Use case: Enforce organizational standards

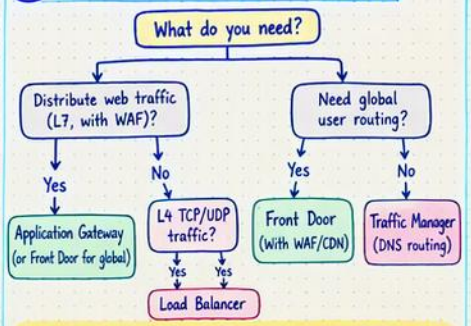
RBAC = who can do it | Policy = what can be done

8 Quick Architecture Flow (Example)



Combine services based on requirements (security, scale, cost, HA)

9 Decision Flow (Which service to use?)



Always consider: security, scale, latency, cost and operational effort

10 Common Azure CLI Commands

Use --help for more options!

```
* Login to Azure                az login
* Show current account          az account show
* Set subscription              az account set --subscription "<subscription-id>"
# List resource groups          az group list -o table
* List VMs (detailed)           az vm list -d -o table
* List VNets                    az network vnet list -o table
* List AKS clusters             az aks list -o table
# Get AKS credentials           az aks get-credentials --resource-group <rg> --name <aks-name>
# Show effective routes for a NIC  az network nic show-effective-route-table \
                                   --resource-group <rg> --name <nic-name>  -o table
* Show effective NSGs for a NIC   az network nic show-effective-nsg \
                                   --resource-group <rg> --name <nic-name> -o table
```

11 Final Interview Answer Checklist

- [x] Start with a clear and concise definition.
- [x] Explain how it works (architecture/flow).
- [x] Mention key features and benefits.
- [x] Give a real-world example.
- [x] Discuss when to use it (and not to use it).
- [x] Cover pricing or SKU considerations (if relevant).
- [x] Mention security, HA/DR and scalability aspects.
- [x] Compare with related services.
- [x] Share troubleshooting tips or common issues.
- [x] End with a key takeaway.

Be Confident Be Clear Be Practical Show Your Thinking All the Best!

It's not just about knowing the answers, but explaining the why!

Azure Today → Better Solutions Tomorrow